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Project Title: From Roads to Classrooms: Educational Infrastructure Investment, Socioeconomic Equity, and Household Travel Behavior in Victoria (2023–2025)

Aims:

This visualisation communicates three connected findings to Victorian policymakers and transport planners:

- Education infrastructure investment broadly tracks where households are located across Victoria's LGAs
- Higher investment tiers correlate with greater car dependency, not active or public transport uptake
- Neighbourhood disadvantage — not investment level alone — is the stronger predictor of how households travel

The goal is to show where government spending and community transport need are aligned, and where they are not.

Motivation:

Victoria spends billions annually on regional infrastructure — but investment alone does not guarantee equity. If funding concentrates in already well-connected areas, or if economically disadvantaged households cannot access viable transport options, the gap between government intention and community outcome widens. This project combines VISTA 2024–25 travel survey data, the 2023–24 State Budget Infrastructure Investment Map, and ABS SEIFA 2021 indexes to assess whether infrastructure decisions are spatially and socially aligned with real household travel demand — and what that means for equity in daily mobility across Victoria.

IDEAS

01

Stacked Bar Chart

02

Grouped Bar Chart

03

Connected Dot Plot

04

Violin + Box Plot

05

Scatter + Trend Line

06

Lollipop Chart

07

Slope Chart

08

Diverging Bar Chart

09

Heatmap

10

Matrix / Table

11

Small Multiples

12

Ranked Dot Plot

13

Scroll Narrative

14

Tab Dashboard

15

Stepper Layout

16

Bump Chart

17

Static Log Page

18

Dropdown Filter

19

Range Slider

20

Tooltip

21

Radio Button

22

Toggle

23

Linked Highlight

24

Headline Numbers

25

Question Decomp.

26

Data Coverage Gap Callout

FILTER

#9 → Heatmap: poor at differentiating small % change across cells.

#10 → White-cell table: individual unit encoding becomes unreadable.

#2 → Toggle: interaction complexity without clear payoff.

#17 → Static log page: needs interactivity to be meaningful.

#6 → Lollipop: redundant with bar + dot plot combination.

#8 → Diverging bar: realistic neutral midpoint may not exist in this data.

#13 → Scroll narrative: JavaScript dependency — excluded per course constraints.

#23 → Linked highlight: patterns matter more than individual row comparison.

CATEGORISE

Ideas that Set Context

#24, #25, #12

Orients the reader before analysis. Establishes scale, frames questions, shows what is most prominent.

Ideas that Frame the Narrative

#25, #26

Tells the reader what data can and cannot show. #26 serves double duty — contextualises entry point and frames the whole narrative.

Ideas that Show Comparisons

#1, #2, #3, #5

Lets the reader see differences between groups or across an ordered axis.

Ideas that Show Distribution & Variability

#4, #6

Reveals spread and shape behind summary stats. #11 can serve as companion across subgroups and as a layout device.

Ideas that Give Reader Control

#18, #19, #20, #21

Interactive controls. Augment charts, not standalone. Risk: complexity without payoff for a policymaker audience.

COMBINE & REFINE

Design A — Sequential Stepper

Fully author-driven. Fixed path Q1→Q2→Q3. Each step: one chart + narrative text panel alongside.

Design B — Dashboard

Fully reader-driven. All panels visible at once. Dropdown + slider for filtering.

Design C — Tabbed Layout

Reader-driven via tabs. One tab per research question. Dropdown filters within each tab.

QUESTIONS

Q1

Does any design give a policymaker a clear sense of what action to take?

Design A ends at a clear policy implication. B and C leave the reader to draw their own conclusions, risking unresolved questions for a non-expert audience.

Q2

How much narrative text does a policymaker need alongside each chart?

Design A allocates explicit text panels per chart. B and C prioritise chart real estate over narration — risky for a policymaker who needs framing to act.

Q3

If a reader enters mid-way, can they understand without reading earlier sections?

Design B handles this best — each panel is self-contained. Design A breaks without sequential reading. C partially handles it via tab labels.

Q4

Does Design A's fixed sequence risk losing a reader interested in only one question?

Yes, but acceptable — policymakers need the full picture to act. Author-controlled narrative is appropriate for this audience.

Q5

Does Design B's simultaneous visibility create cognitive overload?

Yes. Showing all three questions at once without a guided entry overwhelms a non-expert reader. Works for analysts, not policymakers.

Q6

Does Design C's tab structure communicate a single connected argument?

Not clearly. Tabs imply independent content. A reader can close after one tab and miss the full argument. Linking narrative between tabs is needed.

Q7

How do we communicate the structural data gaps without it looking like incomplete analysis?

Frame the 29-LGA coverage gap as a policy-relevant finding: VISTA does not survey regional Victoria, meaning policymakers currently lack travel behaviour evidence to evaluate investment in those areas. Present as a prominent callout, not a limitation footnote — the gap itself is part of the story.

INFORMATION

Title: Design A — Sequential Stepper | **Sheet No:** Sheet #2

Author: Bharath Arun Gandhimani | **Date:** 12/05/2026

Description: Fully author-driven interactive narrative visualisation communicating findings about educational infrastructure investment, socioeconomic equity, and household travel behaviour in Victoria. Targeted at policymakers and transport planners. Reader progresses through a fixed 7-step sequence — one analytical layer per step.

DESIGN RATIONALE

The stepper enforces a deliberate reading order — investment geography is established first before socioeconomic or behavioural patterns are introduced. This prevents a policymaker from jumping to conclusions without the geographic context that frames the entire argument. Chart types (choropleth, slope chart, lollipop) are drawn from Sheet 1 Ideas #7, #6, and the geographic/map category respectively. No filters or dropdowns — narrative control stays with the author.

LAYOUT

1

2

3

4

5

6

7

[F] Step 4 of 7 — Mode Share Shift Across Investment Tiers

[D] Chart Zone (60% content)

Low Inv.

High Inv.

PV 56%

PT 34%

Act 10%

PV 74%

PT 18%

Act 8%

Low Inv. LGAs

PT share: 34%

Private Vehicle

Public Transport

Active

[C] ← Previous

[B] Next →

[A] Step Indicator

[B] Next Button

[C] Previous Button

[D] Chart Zone

[E] Narrative Text

[F] Step Heading

Step 1 — Title Frame

Step 1

Title / Framing Question

Step 2 — Choropleth

Step 2

Choropleth — Investment by LGA

Step 3 — Lollipop

Step 3

Lollipop — LGA Project Count

Step 4 — Slope — Active

Step 4 (Active)

Slope Chart — Mode Share Shift

Step 5 — Grouped Bar

Step 5

Grouped Bar — Vehicle Ownership

Step 6 — Scatter

Step 6

Scatter — IRSD vs Mode Share

Step 7 — Closing

Step 7

Closing — 29-LGA Gap + Policy

OPERATIONS

[B] Click "Next →"

⇒

Content in [D] and [E] fades out. Next step's chart and narrative text fade in. Step indicator [A] advances — current circle fills solid grey (visited), next circle activates black. On the final step the Next button is replaced by a "Restart" option.

[D] Hover on chart element

⇒

A tooltip appears showing exact values — percentage, count, LGA name. This is the only interactivity within a step. No filters, no dropdowns, no view changes are available.

[C] Click "← Previous"

⇒

Returns to the prior step. [A] updates to reflect the previous active step. [C] is hidden when the reader is on Step 1 — there is no "before" the opening frame.

[A] Step circle — state logic

⇒

Visited circles show filled grey. Active circle shows filled black. Unvisited future circles show white with light grey border and are unclickable. State maintained for the full session.

[A] Click a visited step circle

⇒

Jumps directly to that completed step for review. Unvisited future circles remain greyed out and unclickable — the reader cannot skip ahead to content they have not yet reached.

Step 7 — Final step reached

⇒

The 29-LGA gap callout and policy implication text are revealed. Data source links and a "Restart" button appear. The [B] Next button is hidden. Reader has completed the full narrative arc.

FOCUS / PARTI — STEP TRANSITION MECHANISM

Indicator State: N active

N-1

N

N+1

Chart — Step N

Narrative Text

Click Next

⇒

Fade out / Fade in

Indicator State: N+1 active

N-1

N

N+1

Chart — Step N+1

Narrative Text

The closing step only makes sense once all prior evidence has been absorbed — this is what separates Design A from B and C. Chart types per step: Step 2 → Choropleth (geographic category, Sheet 1); Step 3 → Lollipop (Sheet 1 Idea #6); Step 4 → Slope Chart (Sheet 1 Idea #7).

Step Circle States

• Visited — filled grey

• Active — filled black

○ Locked — white, unclickable

Label Reference

[A] Step Indicator

[B] Next Button

[C] Previous Button

[D] Chart Zone

[E] Narrative Text

[F] Step Heading

DISCUSSION

Pros

- Sees evidence in exact order as intended — no confusion about where to start.
- One chart, one explanation per step keeps things simple and focused for a non-technical policymaker audience.

Cons

- Cannot compare two charts side by side — reader must go back and forth between visited steps.

- Opens with geographic framing (choropleth). Closes with what remains unknown (29-LGA gap). Clean narrative arc.
- Works well with non-technical audiences like policymakers who need framing before data.
- Easy to implement in R Shiny using a reactive step counter with conditional panel rendering.
- Hover tooltips as the sole interactivity keeps implementation scope controlled and avoids cognitive overload.

- Feels passive — no filters, no exploration, just clicking Next. Transport planners who are data-savvy may find it too hand-holdy.
- Reader may lose interest if any single step does not feel immediately relevant to their concern.
- Readers who already understand the context must still sit through the opening framing step.
- Step count of 7 is on the upper boundary — risk of fatigue before reaching the policy implication in Step 7.

Title: Design B — Question-Awer Dashboard | **Sheet No:** Sheet #3
Author: Bharath Arun Gandhimani | **Date:** 13/05/2026
Description: Fully reader-driven dashboard communicating findings about educational infrastructure investment, socioeconomic equity, and household travel behaviour in Victoria to policymakers. Three chart panels are visible simultaneously — each poses an explicit question and answers it with a distinct encoding. A left-side filter panel scopes each filter to specific panels only.

Each panel uses a different encoding matched to its analytical task: scatter plot (bivariate investment-vs-household relationship), violin plots (distribution shape across investment tiers), and small multiples (faceted mode share by IRSD decile). Chart types drawn from Sheet 1 Ideas #5 (scatter), #4 (violin), #11 (small multiples). Filters are intentionally scoped — [B1] affects [C] only, [B2] affects [E] only, [B3] affects [C] and [E] — to prevent unintended cross-panel interference. [D] is deliberately unfiltered so the reader always has a stable distributional reference.

[A] Victorian Education Investment & Travel Equity Data Source ↗

[B] Filter Panel

[B1] Trip Purpose
 Work / Education / Both
 → updates [C] only

[B2] IRSD Range Slider
 Decile 2 ——— 10
 → updates [E] only

[B3] Investment Tier

☒ None
☒ Low
☒ Medium
☒ High

(All checked by default)
 → updates [C] + [E]
 [D] unaffected by any filter

Filter Scope
 B1 → [C] only
 B2 → [E] only
 B3 → [C] + [E]
 [D] unaffected by any filter

⚠ [G] VISTA Coverage Gap
 29 LGAs: investment recorded,
 no VISTA survey data

[C] Does investment follow households? Filters: B1 — Purpose | B3 — Tier

VISTA Households

State Budget Investment per LGA (\$M)

None Low Medium High

Casey — High Tier
\$48M | 12,400 HH

[D] Does investment relate to travel behaviour? No filter applied

Trip Duration (min)

None Low Medium High

Median IQR

Hover: tier, n trips, median, IQR

[E] Who actually travels differently? — Mode Share by IRSD Decile Group Filters: B2 — IRSD | B3 — Tier

(Lowest) Deciles 1-2 Deciles 3-4 Deciles 5-6 Deciles 7-8 Deciles 9-10 (Highest)

Mode Share (%)

PV PT Act

Private Vehicle (PV) Public Transport (PT) Active Travel (Act)

Hover: IRSD group, mode, share %, investment tier, LGA count

[B1] Select Trip Purpose dropdown → [C] scatter updates to trips matching selected purpose (Work / Education / Both). [D] and [E] unchanged — purpose-level breakdown not available at LGA-level for violin or faceted mode share.	[B2] Drag IRSD Range Slider → [E] small multiples grey out facets outside selected decile range — highlights relevant band without removing context. [C] and [D] unchanged.	[B3] Uncheck Investment Tier → [C] removes points for unchecked tier and re-draws trend line. [E] removes bars for that tier within each facet. [D] unaffected — violin shows full distribution regardless of tier.
Hover on point in [C] scatter → Tooltip: LGA name, tier, spend (\$M), household count, trip purpose. Hovered point highlighted; all others dim to 30% opacity.	Hover on violin body in [D] → Tooltip: tier label, median duration (min), IQR range, n trips. Median and IQR markers always visible statically — hover supplements, does not replace.	Hover on bar in [E] small multiples → Tooltip: IRSD decile group, mode type, share (%), n LGAs in facet. Same mode bar highlights across all visible facets simultaneously for cross-facet comparison.

[B] Filter Panel
B1 B2 B3
B1: Trip Purpose
B2: IRSD Range
B3: Investment Tier

[C] Scatter Plot
Invest. vs Households
Affected: B1 + B3

[D] Violin Plots
Trip Duration × Tier
No filter applied

[E] Small Multiples
Mode Share × IRSD Decile
Affected: B2 + B3

Parti note
3 questions → 3 encodings
Scatter: bivariate relationship
Violin: distribution shape
Small mult: faceted comparison

[D] unfiltered — stable distributional reference
[E] not affected by B1

Panel Summary
[C] Scatter: X = Budget spend/LGA, Y = VISTA HH, Colour = Tier
[D] Violin: X = Tier (4 groups), Y = Trip duration (min), Overlay = median+IQR
[E] Sm. Multiples: Facet = IRSD decile (5), X = Mode, Y = Share (%)

Label Reference
[A] Title Bar [B] Filter Panel
[B1] Trip Purpose Dropdown
[B2] IRSD Range Slider
[B3] Investment Tier Checkboxes
[C] Scatter Plot Panel
[D] Violin Plots Panel
[E] Small Multiples Panel
[G] VISTA Coverage Gap Callout

Three questions, three completely different encodings — no chart type is reused across panels. Filter scope is asymmetric by design: each filter targets only the panel where it is analytically meaningful. [D] is permanently unfiltered to give readers a stable distributional baseline regardless of how they interact with [B1] or [B3].

Pros	Cons
- A policymaker can navigate directly to the question most relevant to their brief — no compulsory sequence to sit through before reaching the panel they care about. - Each panel uses a distinct encoding matched to its analytical task: scatter for bivariate relationship, violin for distribution shape, small multiples for faceted comparison. No chart type is repeated across panels — each question gets its own visual language. - Asymmetric filter scoping prevents cross-panel noise — each filter only acts where it is analytically valid, reducing reader confusion from unexpected changes. [D] violin plots are permanently unfiltered, giving the reader a stable distributional reference that does not change as they experiment with [B1] or [B3] — useful as an anchor point. - Small multiples in [E] let the IRSD decile gradient emerge visually across all five facets simultaneously without any interaction needed — the pattern is visible at a glance. - R Shiny implementation is straightforward: three independent renderPlotly() outputs, each reactive only to its scoped inputs, avoiding unnecessary re-renders.	- All three panels are visible at once with no guided entry point — a policymaker unfamiliar with the topic may not know where to start and may engage only one panel, missing the full argument. - There is no narrative thread connecting the three questions. A reader who engages only with [C] or [D] misses the socioeconomic equity dimension that [E] communicates. - The VISTA gap callout [G] sits at the bottom of the filter panel and is easy to overlook for a reader whose attention is drawn immediately to the larger chart panels on the right. - Asymmetric scoping (B1 → [C] only, B2 → [E] only, B3 → [C] + [E]) requires orientation time — a first-time reader may apply a filter and be confused when not all panels update. [C] and [E] both involve investment tier as a variable but from different angles — without accompanying explanatory text, this could be perceived as redundant or contradictory rather than complementary. - Unlike Design A (Sheet 2), there is no closing policy implication step — the reader must draw their own conclusions from the three panels, which may be unreliable for a non-expert audience.

INFORMATION

Title: Design C — Martini Glass | **Sheet No:** Sheet #4

Author: Bharath Arun Gandhimani | **Date:** 13/05/2026

Description: Interactive narrative visualisation communicating findings about educational infrastructure investment, socioeconomic equity and household travel behaviour in Victoria to policymakers and transport planners. A two-state structure: State A delivers a guided scroll through four sequential narrative panels [P1–P4], each pairing a bold headline claim with a small embedded chart. A [D] button transitions the reader to State B, where filters [F1–F4] drive a live chart zone [G3] for open exploration.

DESIGN RATIONALE

The martini glass structure gives policymakers the narrative framing they need before exploration — author controls the story entry, reader controls the exit. Chart types are chosen to be fully distinct from Sheets 2 and 3: **dot strip plot** (investment dispersion across LGAs, Q3), **bump chart** (mode share rank shift across investment tiers, Q1), and **connected dot plot with 95% CI error bars** (vehicle ownership by income band, Q2). [F1] Metric dropdown switches the chart type in [G3] — a mode share metric renders a connected dot plot, a duration metric renders a violin+box plot, a vehicle ownership metric renders a connected dot plot with CI bars.

LAYOUT

[A] State A — Guided Scroll

[P1] Q3: Investment broadly follows households

Dot Strip — LGA Investment

[P2] Q1: Higher investment LGAs show more car use

Bump — Mode Rank × Tier

[P3] Q2: Wealthier neighbourhoods own more vehicles

Connected Dot + CI

[P4] Now explore the evidence yourself

[D] Jump to Explore → [H] Source ↗

[B] State B — Open Exploration

[G3] Chart Renders Here

chart title updates per [F1]

Mode Share (%)

80% 60% 40% 20%

← Back to narrative → Neighbourhood Ranking (low → high) →

● Private Vehicle ● Public Transport ● Active Travel

Filters Active

[F1] Metric: Mode Share → changes chart type in [G3]

[F2] Trip Purpose: All (Work + Edu) → re-renders [G3]

[F3] Investment Tier: None, Low, Medium, High → compares with other filters

[F4] IRSD Range: Decile 2 — 10 → removes/greys points outside range

Hover on [G3] → value, group label, ranking, tier, n trips

⚠ VISTA Coverage Gap: 29 LGAs: investment recorded, no VISTA data. Shown as grey stripes in [G3] when relevant.

[A] State A Guided Scroll [B] State B Open Exploration [D] Jump to Explore [F1] Metric [F2] Trip Purpose [F3] Tier [F4] IRSD Range [G3] Chart Zone [H] Source

OPERATIONS

[A] Scroll / natural read through [P1–P3]

⇒ Reader moves through three authored panels in sequence. Each panel is a fixed layout: chart thumbnail (left) + headline claim + narrative text (right). No interaction required — author controls the evidence order. [P4] bridge panel appears at the bottom of the scroll.

[D] Click "Jump to Explore"

⇒ Transitions from State A to State B. Left narrative column [P1–P3] is replaced by the filter panel [F1–F4]. Filters activate from greyed state. [G3] renders with default settings: F1 = Mode Share, F2 = All, F3 = All tiers checked, F4 = full IRSD range. A "← Back to narrative" link stays visible.

[F1] Change Metric dropdown

⇒ [G3] chart type changes: Mode Share → connected dot plot; Trip Duration → violin + box plot; Vehicle Ownership → connected dot plot with 95% CI error bars. Chart title in [G3] updates accordingly. All other active filters [F2–F4] remain applied.

[F2] Change Trip Purpose dropdown

⇒ [G3] re-renders for selected purpose subset (Work / Education / Both). Compares with [F1] and [F3] simultaneously. Chart title updates to reflect selected purpose.

[F3] Change Investment Tier checkboxes

⇒ [G3] re-renders for selected tier(s) only. Unchecked tiers are removed from view. Chart title updates. Compounds with other active filters [F1] and [F2].

[F4] Drag IRSD Range slider

⇒ Data points outside selected IRSD decile range are removed or greyed out in [G3]. Compounds with other filters. Hover on any point in [G3] → tooltip: value, group label, ranking, tier, n trips. [H] data source link always accessible.

FOCUS / PARTI — TWO-STATE TRANSITION

State A — Guided Scroll

[P1] Dot Strip — Investment/LGA

[P2] Bump — Mode Rank × Tier

[P3] Connected Dot+CI — Vehicle Own.

[P4] Bridge: "Explore the data yourself"

Click [D] one-way →

Filters unlock from grey state — Back always available

State B — Open Exploration

[F1] Metric → changes chart type in [G3]

[F2] Trip Purpose → re-renders [G3]

[F3] Invest. Tier → filter [G3] points

[F4] IRSD Range → grey/remove points

Filters have separate drop boxes

Hover [G3] → tooltip always on

Parti Note

Narrative first, exploration second — author decides what evidence matters, reader decides depth.

3 chart types, none repeated from Sheets 2/3

[F1] switches type in [G3]

Chart Type per [F1] Metric

Mode Share → Connected dot plot

Trip Duration → Violin + box plot

Vehicle Ownership → Connected dot + 95% CI

Label Reference

[A] State A Guided Scroll

[B] State B Open Exploration

[D] Jump to Explore button

[F1] Metric dropdown

[F2] Trip Purpose dropdown

[F3] Investment Tier checkboxes

[F4] IRSD Range slider

[G3] Chart Render Zone

[H] Data Source link

[P1–P4] Narrative panels

State A enforces the narrative arc: investment geography (Q3) → mode share consequence (Q1) → socioeconomic mechanism (Q2). State B hands control to a reader who has already absorbed the argument. The [D] transition is one-way by default; "← Back to narrative" restores State A if the reader needs context again. Chart types across all three panels are entirely distinct from Sheets 2 and 3 — dot strip plot, bump chart, and connected dot plot with 95% CI are used here for the first time in the FDS sequence.

DISCUSSION

Pros

- Policymakers receive a structured evidence briefing before controls are handed over — they understand the argument before they can challenge it.
- Small, paired chart-text accumulation in [P1–P3] is cognitively natural: one claim, one chart, one reading at a time.
- [D] gives immediate access to experienced policymakers wanting to explore the data without sitting through the full narrative.
- [P4] makes it clear to readers that they have finished reading and are entering a different mode — reducing the ambiguity of the transition.
- "← Back to narrative" link enables users to come back to the guided mode without losing exploration state.
- Three chart types in [P1–P3] are analytically distinct and none are repeated from Sheets 2 or 3 — guarantees genuine FDS design space coverage.

Cons

- For experienced policymakers, the static narrative guided mode may feel slow — they may jump to [D] immediately, bypassing the intended framing.
- Placing the exploration zone on the right side in State B may make the reader think there is an exploration mode before even reading the guided version.
- Four stacked narrative panels make the left column long — readers on small screens face significant scrolling before reaching the [D] bridge panel and exploration mode.
- [F1] switching the chart type in [G3] changes the visual encoding significantly — a reader who has not read the panel descriptions may be confused by the sudden appearance of a violin plot rather than a dot plot.

INFORMATION

Title: Realization — Victoria Is Building Schools. But Not Transport. | **Sheet No:** Sheet #5

Author: Bharath Arun Gandhimani | **Date:** 17/05/2026

Description: Interactive narrative visualisation communicating findings about educational infrastructure investment, socioeconomic equity and household travel behaviour in Victoria to policymakers and transport planners. Martini glass structure: persistent two-column layout. Left column is a fixed three-part authored narrative scroll (State A). Right column activates as an LGA Explorer (State B) when reader signals readiness. No filters. Hover tooltips are the sole interactivity in the narrative zone.

DESIGN RATIONALE & LINEAGE

Synthesises across all prior sheets. Martini glass architecture from Sheet 4. Dot strip chart [P1] adapted from Sheet 3 [C] scatter. Connected line chart [P2] adapted from Sheet 4 [P2] slope chart — redesigned with a consistent investment tier x-axis for legibility. Connected dot plot [P3] from Sheet 4 [P3]. Right-zone filters [F1-F4] from Sheet 4 **removed** — filter-driven exploration places analytical burden on the reader inappropriate for a policymaker audience. Replaced by a single LGA dropdown that personalises an already-delivered finding to a specific suburb. 29-LGA gap callout from Sheet 1 Idea #26, promoted to a persistent right-zone element visible in both states.

